

Team Reflection

Project #: 7

Project Title: Hexapod Robot

Category: A

Team Members: Prachi Gopani | H021

Hrishikesh Gudekar | H022

Daksh Hariya | H023

Date: 19/04/2026

Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

The hardest challenge was deriving and integrating the forward and inverse kinematics of the hexapod legs. While solving for a single leg was manageable, extending it to all six legs with correct orientation and symmetry caused major errors in positioning.

We resolved this by:

- Defining consistent coordinate frames using generalized parameters.
- Solving kinematics for one leg first.
- Replicating it across all six legs with proper angular offsets.

Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?

The most challenging was CO4: Applying constitutive laws to analyze dynamic systems of particles and rigid bodies.

This was difficult because:

- It required moving beyond ideal kinematics to real-world behavior of joints and links.
- Needed combining multiple concepts like motion, constraints, and system response.
- Transforming theoretical workings into working model was complex.

Q3: What would you do differently if you started the project over?

If we restarted the project:

- We would begin simulation earlier instead of completing full derivations first.
- Not testing it on PyBullet in the earlier stage caused more problem as the leg geometry was correct but at the time of simulation the bot could not stand in proper stance.
- Follow a modular approach (one leg -> gait -> full robot).
- Integrate force and motion analysis together to reduce rework.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

This project taught us things that textbook problems don't:

- Real-world systems require iteration and debugging constantly.
- Small mistakes in modeling can cause large errors in results.
- Importance of simulation tools for validation.

- Understanding how multiple concepts work together in a system, not separately.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

Work was divided as:

- One member handled kinematics derivation.
- One worked on simulation and coding.
- One focused on integration, debugging, and documentation.

Despite this division:

- We regularly discussed progress so everyone understood the full model.
- Cross-checking ensured accuracy and shared understanding.

Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?

We used AI tools like **ChatGPT/Claude/GLM(Z.ai)** for:

- Understanding complex concepts like inverse kinematics and modeling.
- Writing Python/PyBullet code from MATLAB code.
- Structuring explanations and documentation.

We did NOT use AI for:

- Deriving problem, it's mathematical modeling concept and FBDs.
- Simulation validation.
- Core conceptual learning.